A065-v0.4.1 | Arc A770 3D Euler numerical audit

Release category: NUMERICAL SCREENING ONLY

This report records device-executed FFT/RK4 tests only when an actual Intel Arc A770 Level Zero device is identified. Missing hardware produces NOT_RUN_PREREQUISITE, never PASS. Synthetic Gaussian-initialized Euler data do not establish physical PKLSA, Casimir, gravity, or independently observed pressure.

Pre-registered inputs

A003 source geometries	torus_3_1_direct / torus_3_1_hoste
Grid ladder	48^3 / 64^3 / 96^3
Numerical cube	L = 12 periodic
Initial vortex core	Gaussian a = 0.55 (assumed)
Circulation / density	Gamma = 1; rho = 1
Euler integrator	Fourier projected rotational RK4; 4 steps, T = 0.02
GPU	SYCL Level Zero / dpnp FFT FP32
Reference	NumPy FFT FP64, same initial data
Precision warning	Previous DD32 Biot-Savart parity is NOT FFT/Euler parity

New gates G43-G48

Gate	Result	Interpretation
G43	PASS	Explicit Arc A770 Level Zero GPU queue; no fallback
G44	PASS	Same operator/initial data GPU FP32 vs independent NumPy FP64 Taylor-Green (not a PKLSA physical test)
G45	PASS	A003 trajectories GPU FP32 vs NumPy FP64 at frozen 48/64/96
G46	PASS	Periodically reconstructed Euler pressure GPU FP32 vs CPU FP64; no independent physical pressure measurement
G47	PASS	Full three-resolution dual-backend result/manifest with explicit A770 identity; source-hash fingerprint required
G48	UNRESOLVED	Gaussian initialization from A003 geometry and dependent numerical pressure cannot qualify native PKLSA physics

Detected GPU: Intel(R) Arc(TM) A770 Graphics

A003 torus_3_1_direct

N=48: relative L2 u=1.56376e-07; omega=3.23805e-07; numerical p=2.58067e-07; div RMS=9.01681e-10

N=64: relative L2 u=1.44598e-07; omega=4.35921e-07; numerical p=1.96432e-07; div RMS=1.06149e-09

N=96: relative L2 u=1.55555e-07; omega=6.04953e-07; numerical p=2.29762e-07; div RMS=1.00053e-09

A003 torus_3_1_hoste

N=48: relative L2 u=1.67059e-07; omega=2.57228e-07; numerical p=2.9269e-07; div RMS=5.65308e-10

N=64: relative L2 u=1.43177e-07; omega=3.10489e-07; numerical p=2.15385e-07; div RMS=6.61501e-10

N=96: relative L2 u=1.59077e-07; omega=4.31402e-07; numerical p=2.53046e-07; div RMS=6.3065e-10

Authority boundary

G48 remains UNRESOLVED. The Euler solver integrates assumed geometry-derived initial vorticity; no external physical transverse-vorticity, force or pressure holdout is supplied. Historical failed gates are unchanged.